CIMAGE
Knowledge . Skill . Success
CIMAGE GROUP OF INSTITUTIONS

HACKATHON PRESENTATION

PlacementAI – Intelligence Engine

AI-first, explainable recruitment for campus placements

PARTICIPANT

Aditya Raj (Solo)

UNIVERSITY

AKU

Aryabhatta Knowledge University

PROGRAM

BCA 2nd Sem

(2025–28)

THE PROBLEM

Hiring Bottlenecks

Keyword Blindness

ATS misses great talent without exact term matches. Example: A highly qualified candidate is rejected because their resume says **'FastAPI'** instead of **'Backend'**.

Unstructured Chaos

Messy PDF and DOCX formats break traditional parsing tools. Critical skills and project experiences are lost during automated data extraction, resulting in incomplete profiles.

Black Box Rejections

Rejected candidates receive zero feedback on why they weren't selected. This lack of transparency leaves students frustrated and without a clear path to improve their skills.

Scalability Gap

During rapid college placement drives, processing 1,000+ applicants simultaneously overwhelms manual evaluation, leading to rushed decisions and missed talent.

OUR SOLUTION

Semantic Intelligence

Neural Parsing

Using LLMs (Gemini/Llama) to robustly convert raw, unstructured resume text into a standardized **"Skill-Experience-Project"** schema.

Vector Search

Implementing Cosine Similarity on text embeddings to find **mathematical meaning matches**, rather than relying on exact keyword comparisons.

Skill Ontology

A custom knowledge graph that intelligently maps relationships between technologies (e.g., understanding that **React** ↔ **Frontend**).

Explainable AI (XAI)

Provides a clear reasoning report for decisions alongside a **Personalized Learning Roadmap** to help candidates bridge their skill gaps.

TECHNICAL STACK

System Architecture

Frontend
React Dashboard

Features real-time Score Ring animations and interactive XAI visualization for feedback.

Backend
FastAPI / Flask

Handles the core NLP extraction pipeline, scoring logic, and API endpoints.

Data Layer
SQLite/PostgreSQL

Relational schema linking Users, Resumes, and Jobs via Foreign Keys.

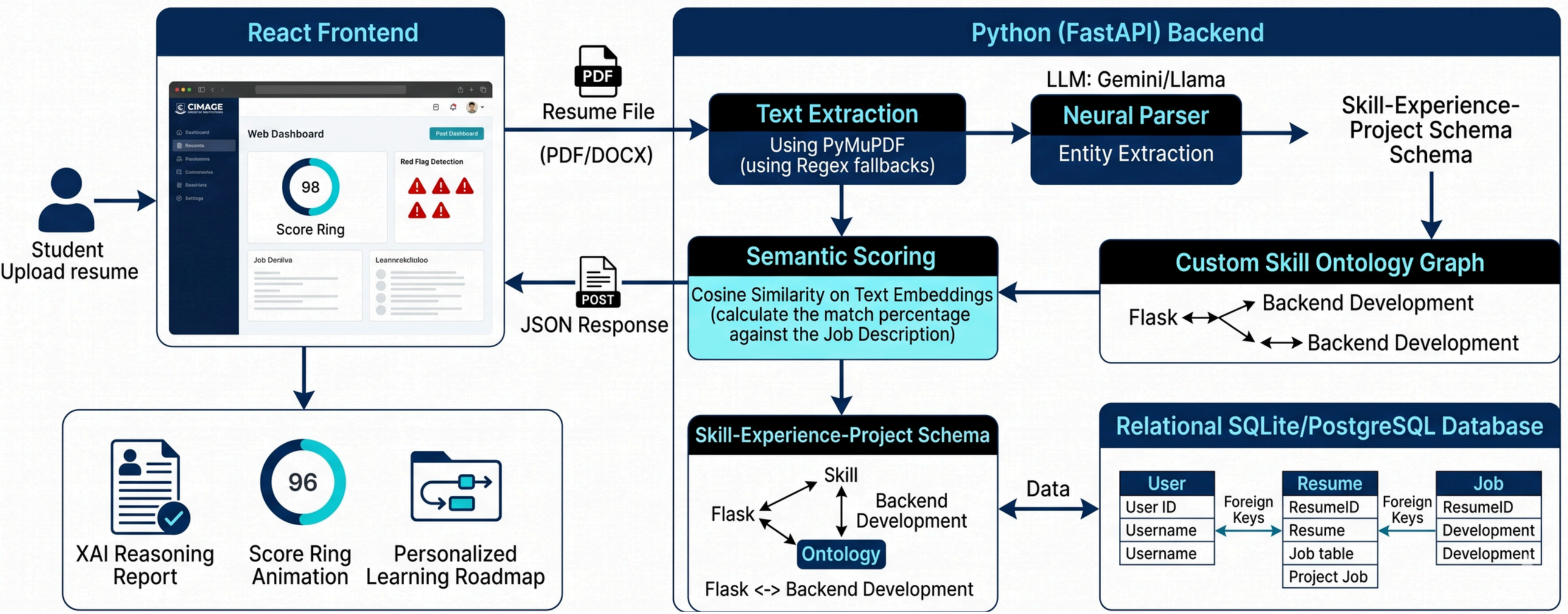
Extraction
PyMuPDF & Regex

Robust text processing pipeline with regex fallbacks for complex document formats.

Performance
High Concurrency

Optimized architecture designed for sub-2 second response times under heavy load.

CIMAGE
Knowledge . Skill . Success
CIMAGE GROUP OF INSTITUTIONS



CONCLUSION

Impact & Vision

Scalability

Ready for 1,000+ concurrent users in a high-pressure college placement environment.

Innovation

Includes "Red Flag" data quality checks and interactive Career Coach insights.

Vision

Democratizing recruitment by focusing on potential and semantic context over rigid keywords.

THANK YOU!

Presenter: **Aditya Raj**

Submission Date: April 19, 2026